

Each question is worth 5 points for a total of 100 points. The Extra Credit problem can get you up to an additional 5 points.

Problem 1

Express the area and perimeter of an equilateral triangle as a function of the triangle's side length x .

Problem 2

In the following, one of $\sin(x)$, $\cos(x)$, and $\tan(x)$ is given. Find the other two if x lies in the specified interval.

a) $\sin(x) = \frac{4}{5}$, $x \in [\frac{\pi}{2}, \pi]$.

b) $\cos(x) = \frac{5}{13}$, $x \in [-\frac{\pi}{2}, 0]$.

Problem 3

Let $h(x) = x^4 - 2x^2 + 6$. Find (a) the open intervals on which the function is increasing and decreasing, and (b) identify the function's local and absolute extreme values, if any, saying where they occur.

Problem 4

Let $f(x) = \sqrt{x+1}$, $g(x) = \frac{1}{x+4}$, and $h(x) = \frac{1}{x}$.

a) Write a formula for $f \circ g \circ h$.

b) Write a formula for $h \circ g \circ f$.

Problem 5

Using the definition of a limit, prove that

$$\lim_{x \rightarrow 3} (2x - 4) = 2$$

Problem 6

Let $f(x) = \sin(x - \sin x)$. Find $\lim_{x \rightarrow \pi} f(x)$ and determine if the function f is continuous at π .
Hint: a function f is continuous at x_0 if and only if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

Problem 7

Find

$$\lim_{x \rightarrow -3} \frac{2 - \sqrt{x^2 - 5}}{x + 3}$$

Problem 8

Find the slope of the function $f(x) = x + 2x^2$ at the point $(1, 3)$. Then find an equation for the line tangent to the graph at the given point.

Problem 9

Find $\frac{dz}{dw}$ where $z = \frac{1}{\sqrt{w^2+w}}$.

Problem 10

Find the first and second derivatives of $y = \frac{-x^3-7}{x}$ with respect to x .

Problem 11

Find $\frac{dr}{d\theta}$ where $r = 4 - \theta^3 \sin(2\theta)$.

Problem 12

Find the slope of $y^2 + x^2 = y^4 - 2x$ at the points $(-2, 1)$ and $(-2, -1)$.

Problem 13

If $x^2 + y^2 = 25$ and $dx/dt = 2$, then what is dy/dt when $x = 3$ and $y = -4$?

Problem 14

Identify the inflection points and local maxima and minima of $y = (x + 2)^3 - 1$. Identify the intervals on which the function is concave up and concave down.

Problem 15

You have been asked to design a half-liter (500 cm^3) can shaped like a right circular cylinder. What dimensions will use the least material? Hint: you're trying to minimize the surface area of the can.

Problem 16

Use Newton's method to get a decimal approximation of $\sqrt{7}$ from $f(x) = x^2 - 11$. Start with $x_0 = 3$ and find x_2 . Then calculate $f(x_2)$.

Problem 17

Find an antiderivative for each function (you only need to find a single antiderivative, not a family of functions).

a) $f(x) = x^{-4} + 2x^2 + 3$

b) $f(x) = \frac{5}{\sin(x)}$

c) $f(x) = 3 \csc(2x)$

Problem 18

Suppose that $f'(x) = 2x + 1$ for all x . Find $f(3)$ given

a) $f(0) = 2$

b) $f(1) = 0$

c) $f(0) = 4$

Problem 19

Solve: $\int(\sqrt{x} + 2x^3)dx$.

Problem 20

Solve: $\int(-\tan(x)\sec(x))dx$.

Extra Credit

A 13-ft ladder is leaning against a house when its base starts to slide away. By the time the base is 12 ft from the house, the base is moving at the rate of 5 ft/sec. Find (a) how fast the top of the ladder is sliding down the wall, (b) at what rate the area of the triangle formed by the ladder, wall, and ground is changing, and (c) at what rate the angle between the ladder and the ground is changing.